Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

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Scenario-Based Requirement Analysis
CO - 2
Assessment Tool - 1

Problem Statement

35. Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

Industry Context

During natural disasters and humanitarian crises, governments and relief organizations establish emergency shelters to accommodate displaced populations. Efficient shelter management is essential for ensuring adequate food, medical care, accommodation, and security. AI, GIS, IoT, cloud
computing, and mobile technologies enable real-time coordination of shelter operations and humanitarian assistance.

Problem Statement

Emergency shelters often experience overcrowding, uneven resource distribution, and poor coordination due to limited real-time information. Manual management methods make it difficult to
monitor occupancy, supplies, and emergency requirements. An AI-based shelter management platform should track shelter occupancy, manage food and medical inventories, allocate resources based on demand, monitor volunteer activities, and generate operational dashboards. The system should improve disaster response while ensuring effective humanitarian assistance.

Objectives

1. Monitor shelter occupancy.
2. Manage relief resources digitally.
3. Optimize resource allocation using AI.
4. Track food and medical supplies.
5. Coordinate volunteers efficiently.
6. Generate shelter management reports.
7. Improve emergency response.
8. Enhance humanitarian service delivery.

Expected Outcomes

1. Efficient shelter operations.
2. Better resource utilization.
3. Faster disaster response.
4. Improved humanitarian assistance.
5. Enhanced volunteer coordination.

6. Increased transparency.

7. Better emergency planning.

8. Improved community resilience.

Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

Introduction

1.Purpose

The Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform is designed to improve the management of relief camps during natural disasters and humanitarian emergencies. Governments and relief organizations often face difficulties in monitoring shelter occupancy, distributing essential resources, and coordinating volunteers. This platform uses Artificial Intelligence (AI), cloud computing, and GIS technologies to simplify these activities. The main purpose of this system is to provide a centralized digital solution that enables authorities to manage shelters efficiently, make faster decisions, and improve the quality of humanitarian assistance.

2.Scope of the Document

This Software Requirement Specification (SRS) describes the complete requirements of the proposed system. It explains the problem being addressed, the objectives of the project, system scope, functional and non-functional requirements, actors, use cases, data requirements, interfaces, assumptions, constraints, and validation process. The document acts as a guideline for the design and development of the platform.

Problem Description

Natural disasters such as floods, earthquakes, cyclones, and landslides often force thousands of people to move into temporary shelters. Managing these shelters manually becomes difficult because administrators must continuously monitor the number of people, available food, medicines, and volunteer activities. Manual record keeping is slow and increases the possibility of errors, resulting in overcrowding, resource shortages, and delayed emergency responses.

The proposed AI-based platform addresses these issues by collecting real-time information from shelters and using intelligent algorithms to predict future resource requirements. The system also automates resource allocation, volunteer coordination, and report generation, allowing relief organizations to respond more effectively during emergencies.

Scope of the System

The proposed system provides a centralized platform for managing emergency shelters and relief camps. Shelter administrators can register shelters, monitor occupancy levels, manage food and medical inventories, and assign volunteers to different tasks. Government officials can monitor the status of multiple shelters through a dashboard, while relief organizations can use AI-generated recommendations to allocate resources efficiently.

The platform also stores historical disaster data, generates operational reports, and sends notifications whenever emergency situations or resource shortages occur. Although the system supports shelter management activities, it does not include disaster rescue operations or transportation management.

Functional Requirements

The system provides several important features to improve shelter management. Users must first authenticate themselves before accessing the platform. Administrators can create and manage shelters, while shelter managers can register residents and update occupancy information. The inventory module keeps track of food, medicines, drinking water, and other essential supplies available in each shelter.

The AI component analyzes occupancy trends and predicts future demand for resources, allowing administrators to prepare before shortages occur. Volunteers can view assigned tasks and update their progress, while medical staff can maintain treatment records and monitor medicine availability. The system also generates reports that summarize shelter operations and resource utilization.

Non-Functional Requirements

The platform should provide fast and reliable performance even during large-scale disasters. It must support thousands of users simultaneously without affecting response time. Security is an important requirement, so all users should access the system through secure authentication and role-based authorization. Sensitive information must be encrypted to ensure data privacy.

The system should remain available throughout the day with minimal downtime. A user-friendly interface should allow administrators, volunteers, and government officials to use the platform with minimal training. Since disasters may affect multiple regions, the application should also be scalable and easy to maintain as new features are introduced.

Actors and Use Cases

The primary users of the system include the Administrator, Shelter Manager, Volunteer, Medical Officer, Government Officer, and Shelter Residents. Each actor interacts with the system based on their responsibilities.

Administrators manage users, shelters, and reports. Shelter managers update occupancy details, maintain inventories, and coordinate relief activities. Volunteers receive assigned tasks and report their completion status. Medical officers update medicine stock and patient records, while government officials monitor disaster situations using dashboards and reports. Shelter residents can register themselves, receive notifications, and access shelter information.

Use Case Diagram (Text Representation)

```
            Administrator
                  |
 ---------------------------------------
 |        |         |         |
Manage Users  Reports   Dashboard  Shelter Control

          Shelter Manager
                 |
 ---------------------------------------
 |          |           |
Occupancy   Inventory    Resource Allocation
```

```
        Volunteer
            |
     -----------------
     |            |
 View Tasks    Update Status

      Medical Officer
            |
     --------------------
     |              |
Medicine Records     Stock Update

        Shelter Resident
             |
      -------------------
      |             |
 Registration   Notifications
```

Data Requirements

The platform maintains several types of information to support shelter operations. Shelter records contain information such as shelter name, location, capacity, and current occupancy. Resident records store personal information and emergency contact details. Inventory records maintain details of available food, medicines, clothing, and drinking water along with their quantities.

Volunteer information includes personal details, assigned shelter, skills, and working schedule. The system also stores reports, AI prediction results, notification history, and audit logs. Proper database management ensures that this information remains accurate, secure, and easily accessible for decision-making.

External Interface Requirements and System Features

The application provides a responsive web interface that can also be accessed using mobile devices. The dashboard displays shelter occupancy, available resources, volunteer activities, and emergency alerts using charts and graphical reports. The backend communicates with the database through REST APIs and exchanges information in JSON format.

The platform integrates with GIS services for shelter location mapping and notification services for sending alerts through email or SMS. Artificial Intelligence models analyze historical and real-time data to predict resource shortages and recommend better allocation strategies. These integrations improve the efficiency of disaster management operations.

Assumptions and Constraints

The proposed system assumes that shelters have internet connectivity and authorized personnel regularly update occupancy and inventory information. It also assumes that historical disaster data is available to train AI models for prediction.

Some constraints may affect the project implementation. The system depends on cloud infrastructure and reliable network connectivity for real-time updates. Budget limitations may restrict advanced hardware integration such as IoT sensors. The project must also comply with government regulations related to disaster management and data privacy. Since this is an academic project, development and testing must be completed within the available semester duration.

Requirement Validation and Conclusion

The proposed requirements have been carefully reviewed to ensure that they satisfy the needs of all stakeholders, including government agencies, shelter administrators, volunteers, and relief organizations. The requirements are written clearly to reduce ambiguity and avoid unnecessary duplication. Functional and non-functional requirements are consistent with the project objectives and support efficient shelter management.

Future enhancements may include IoT-based occupancy monitoring, multilingual support, predictive disease outbreak analysis, offline synchronization for disaster-affected areas, and blockchain-based resource tracking. These improvements can further increase transparency and operational efficiency.

In conclusion, the **Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform** provides a smart and reliable solution for managing emergency shelters. By combining Artificial Intelligence with modern web technologies, the platform enables better resource utilization, faster emergency response, and improved humanitarian assistance during disaster situations.